

AE03 - Exploring gerrymandering

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Getting started

Load the `tidyverse`, `usdata`, and `sjPlot` packages, and get the `gerrymander` data out of the `usdata` package. View the data using `glimpse`.

```
library(tidyverse)
library(usdata)
library(sjPlot)
library(gtsummary)

gerrymander <- usdata::gerrymander
```

1. Congressional districts per state

Which state has the most congressional districts? The least? How many congressional districts are there in this state? Use functions such as `table`, `tbl_summary` or `count`.

```
table(gerrymander$state)
```

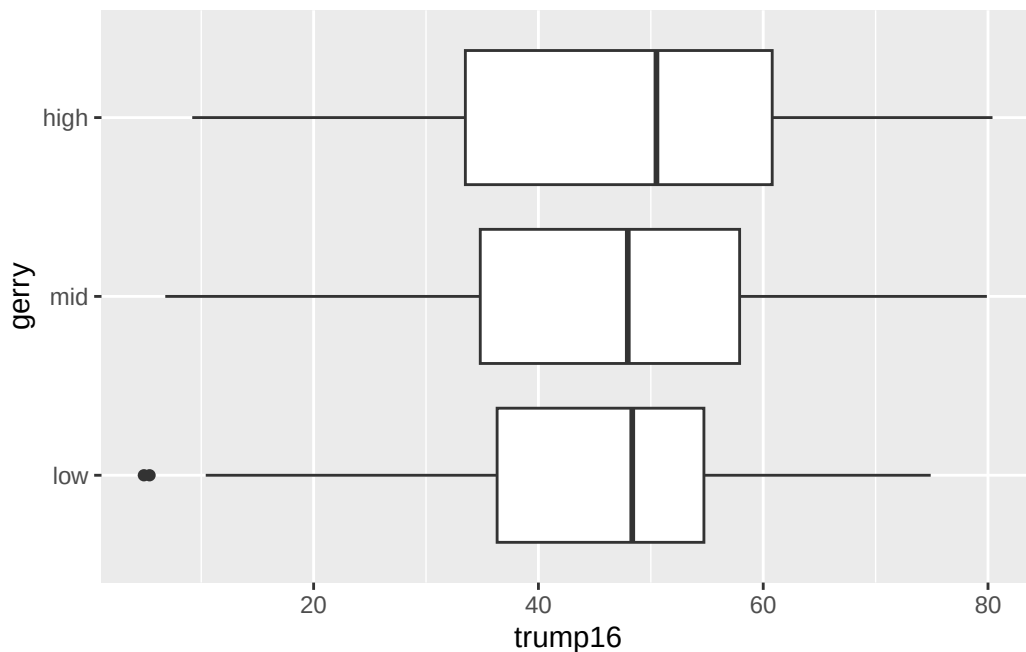
```
AK AL AR AZ CA CO CT DE FL GA HI IA ID IL IN KS KY LA MA MD ME MI MN MO MS MT
  1  7  4  9 53  7  5  1 27 14  2  4  2 18  9  4  6  6  9  8  2 14  8  8  4  1
NC ND NE NH NJ NM NV NY OH OK OR PA RI SC SD TN TX UT VA VT WA WI WV WY
13  1  3  2 12  3  4 27 16  5  5 18  2  7  1  9 36  4 11  1 10  8  3  1
```

California has the highest amount of congressional districts at 53. The least is a tie between Arkansas, Vermont, Wyoming, North Dakota, South Dakota all having 1 congressional districts.

2. Districts at the tails

Make side-by-side box plots of percent of vote received by Trump in the 2016 Presidential Election by prevalence of gerrymandering. Identify any Congressional Districts that are potential outliers — points that fall well outside the whiskers of their box, meaning that district's Trump vote share was unusually high or low compared to other districts with the same gerrymandering prevalence. What information can you share about them?

```
ggplot(gerrymander, aes(x = trump16, y=gerry)) +  
  geom_boxplot()
```



```
gerrymander |> select(district,trump16,gerry) |> filter(gerry=="low" & trump16<=10)
```

```
# A tibble: 2 x 3  
  district trump16 gerry  
  <chr>      <dbl> <fct>  
1 NY-13      5.4 low  
2 NY-15      4.9 low
```

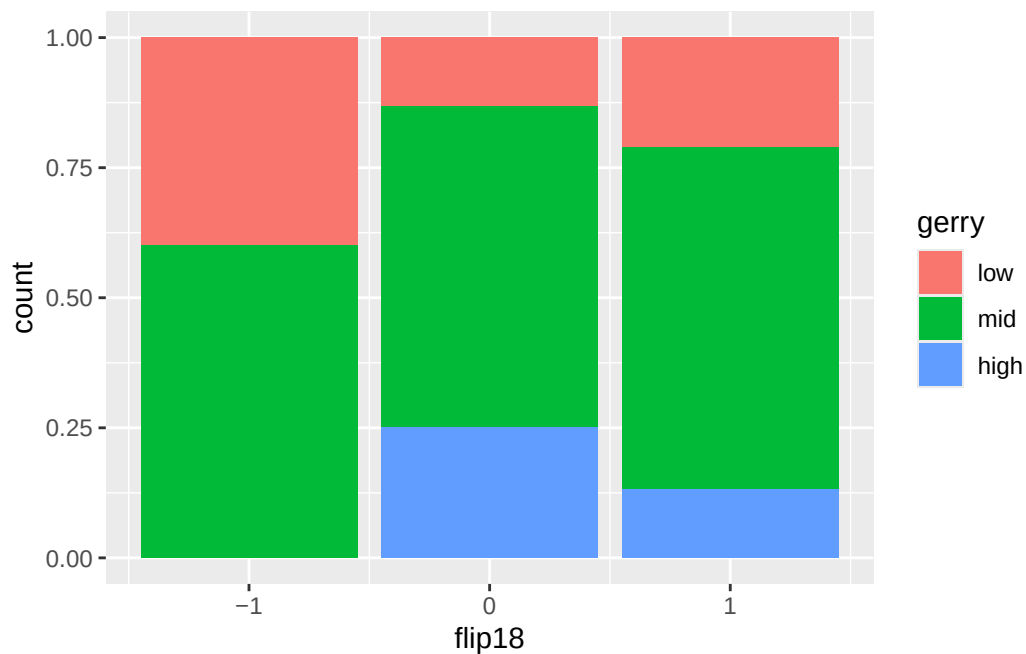
New York districts 13 and 15 have an unnaturally low gerrymandering rate. Ny-13 had a low gerrymandering but a voting for Trump of 5.4% and Ny-15 had a low gerrymandering rate however only 4.9% of people voted for Trump. This could be cause for more investigation.

3. Gerrymandering and flipping

This relationship can be looked at from two directions — does gerrymandering predict a 2018 flip, and does a 2018 flip predict gerrymandering? Support **both** with a visualization and summary statistics. The plot answers “what does the mix look like within each group?”; the table gives the exact proportions the plot is showing. Write 1-2 bullet points about what you noticed from each plot. Use specific numbers in your answer.

Direction 1: Is a Congressional District more likely to have high prevalence of gerrymandering if a Democrat was able to flip the seat in the 2018 election? To compare the breakdown of gerrymandering prevalence within each flipped group, make sure `flip18` is on the x axis and that your percents add up to 100% within each category.

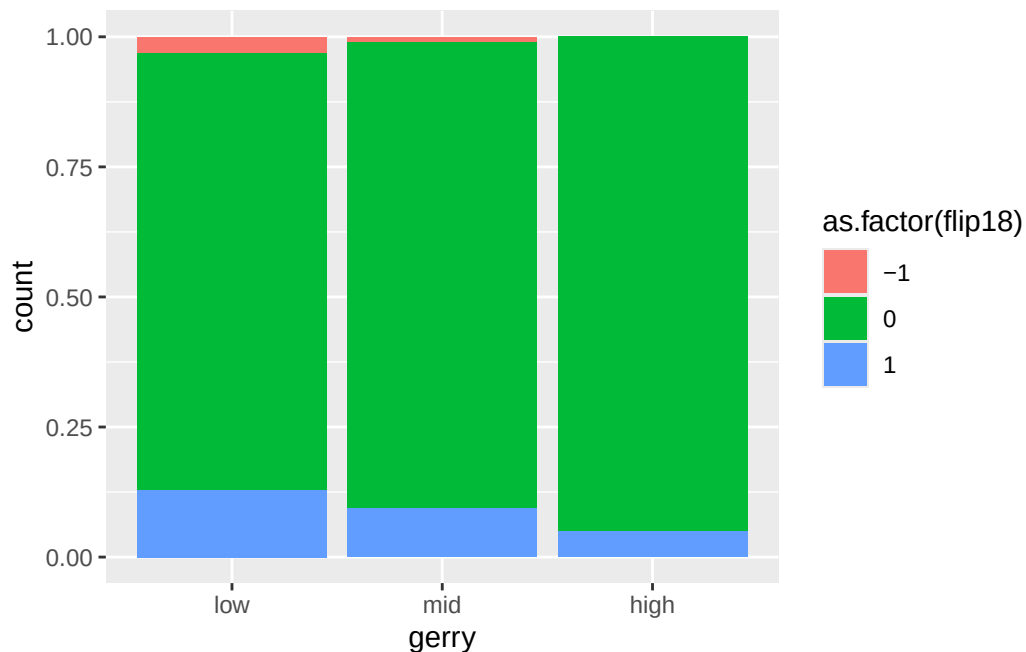
```
ggplot(gerrymander, aes(x = flip18, fill = gerry)) +  
  geom_bar(position = "fill")
```



The states that didn't flip had the highest ratio of gerrymandering overall.

Direction 2: Is a Congressional District more likely to be flipped to a Democratic seat if it has high prevalence of gerrymandering, or low prevalence? Same two variables, just flipped — so now `gerry` is on the x-axis. Again, make sure your percents add up to 100% within each category.

```
ggplot(gerrymander, aes(x = gerry, fill = as.factor(flip18))) +
  geom_bar(position = "fill")
```



A majority of the states didn't flip seats. Especially since the rates for each gerrymander rates were fairly consistent through out the different gerrymandering weights.

4. Gerrymandering and presidential margin

Using `mutate`, create a new variable, `margin16`, measuring how many percentage points Clinton won or lost by in each district (`clinton16 - trump16`). Does gerrymandering prevalence relate to how lopsided the 2016 presidential race was in that district? Report the min, mean, and max margin by `gerry` level, then visualize the full distribution with a violin + boxplot combo. Describe the shape, center, and spread of each group.

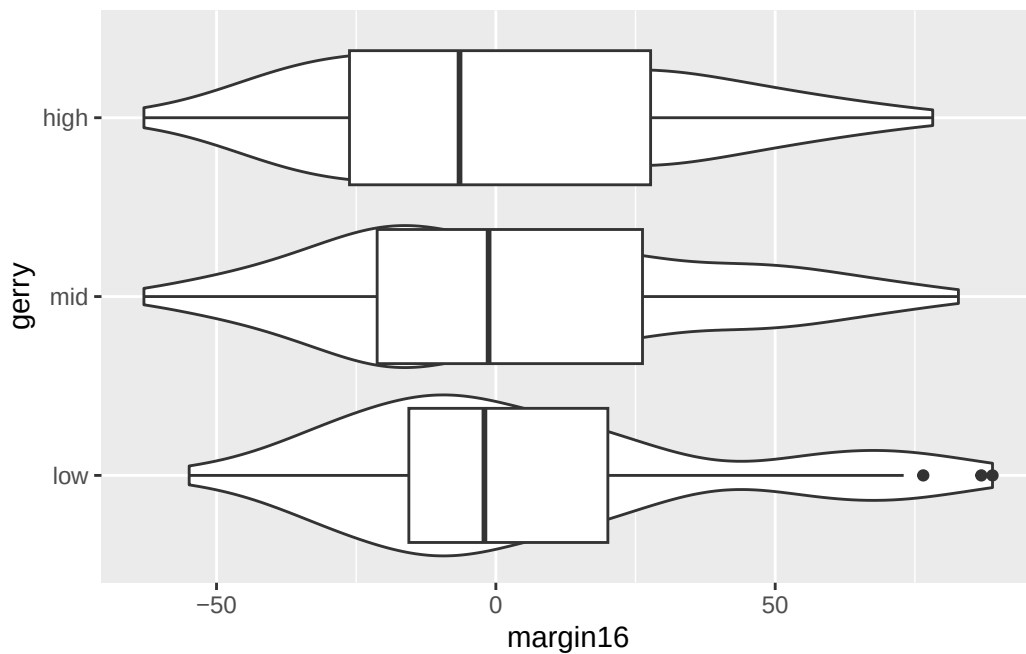
```
gerrymander %>%
  mutate(margin16 = clinton16 - trump16) %>%
  select(gerry, clinton16, trump16, margin16) %>%
  arrange(margin16) %>%
  group_by(gerry) %>%
  summarise(min_margin16 = min(margin16),
            mean_margin16 = mean(margin16),
            median_margin16 = median(margin16),
```

```
max_margin16 = max(margin16)
)
```

```
# A tibble: 3 x 5
```

	gerry	min_margin16	mean_margin16	median_margin16	max_margin16
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>
1	low	-54.9	6.32	-2.05	88.9
2	mid	-63	3.25	-1.3	82.8
3	high	-63	1.34	-6.5	78.2

```
gerrymander %>%
  mutate(margin16 = clinton16 - trump16) %>%
  select(gerry, clinton16, trump16, margin16) %>%
  arrange(margin16) %>%
  group_by(gerry) %>% ggplot(aes(x = gerry, y = margin16)) +
  geom_violin()+geom_boxplot()+coord_flip()
```



Low gerrymandering weights had minimum value -54.9% and a max value of 88.9% when comparing Clinton voting rates to Trump for each district. The low gerrymandering districts also had a median of -2.05% and a mean of 6.32%. The low gerrymandering rate had a right skew, with a wide spread, and centered at 6.32%.

Medium gerrymandering weights had minimum value -63% and a max value of 82.8% when comparing Clinton voting rates to Trump for each district. The medium gerrymandering districts also had a median of -1.3% and a mean of 3.2%. The medium gerrymandering rate had a fairly even spread, was skewed to the right, centered at 3.2%.

High gerrymandering weights had minimum value -63% and a max value of 78.2% when comparing Clinton voting rates to Trump for each district. The high gerrymandering districts also had a median of -6.5% and a mean of 1.34%. The high gerrymandering rate had a fairly even spread, was skewed to the right, centered at 1.34%.